

9	(a)	Fig. 9.2 shows an exponential curve and hence count rate tends towards zero but never becoming zero. So theoretically, complete shielding is not possible.																			
	(b)	(i)	Fig. 9.3 shows a straight line of negative gradient passing through the origin so general form of equation for the line is $Y = -mX$ B1 $\ln(C_X/C_0) = -m \times$ B1 where $-m$ is the gradient of the straight line $(C_X/C_0) = e^{-m \times}$ $C_X = C_0 e^{-m \times}$ where gradient $-m = \text{constant} - \mu$																		
		(ii)	gradient $= -\mu$ M1 $= -3.9 / 8.8$ $\mu = 0.443$ A1																		
	(c)	(i), (ii)	<table><tr><td>material</td><td>μ / cm^{-1}</td><td>$\rho / \text{g cm}^{-3}$</td><td>$\mu_m / \text{cm}^2 \text{g}^{-1}$ B1</td></tr><tr><td>aluminium</td><td>0.095</td><td>2.70</td><td>0.035</td></tr><tr><td>tin</td><td>0.267</td><td>7.28</td><td>0.037</td></tr><tr><td>lead</td><td>.....A0.....</td><td>11.3</td><td>0.039 B1.</td></tr></table> Fig. 9.4			material	μ / cm^{-1}	$\rho / \text{g cm}^{-3}$	$\mu_m / \text{cm}^2 \text{g}^{-1}$ B1	aluminium	0.095	2.70	0.035	tin	0.267	7.28	0.037	lead	 A0	11.3	0.039 B1 .
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	(d)	(i)	average $\mu_m = 1/3 (0.035 + 0.037 + 0.039) = 0.037$ B1 $2.4 \times 10^3 \text{ kg m}^{-3} = 2.4 \text{ g cm}^{-3}$ $\mu = \mu_m \times \rho = 0.037 \times 2.4$ M1 $0.0888 \approx 0.09 \text{ cm}^{-1}$																		
		(ii)	Read off from Fig. 9.2, $C_X/C_0 = 0.16$ B1 Given the relationship, $C_X/C_0 = e^{-\mu x}$ $0.16 = e^{-0.0888 x}$ M1 $\ln(0.16) = -0.0888 x$ $x \approx 20 \text{ cm (approximate) (1 s.f.)}$ A1																		